Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I think that all the concepts from class that we have previously learned will come in handy will completing this project. I expect to use loops, conditions, dataclass, and possibly lists if necessary. At this time I have ideas but I do not know if they will change and adjust once I get to the actual coding of my project. I hope to improve my speed and my use of for statements while doing this project. I tend to take a really long time while coding. I am hoping that by using AI to guide me and write code it will help me understand but also move faster.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the easiest part of the project is going to be putting together all my ideas and then eventually sending out my code. I think this because I do have a true passion for the topic that I chose so I do not think that that aspect of the code is going to be hard. Because I know my plan and have ideas. I also know a lot of people that would love to test my code so I think that will also be helpful when I need to find where my project may break. The hardest part is definitely going to be making the code that does what I want it to do. I hope that I am going to able to get my code to work the way I want it to

and that I wont have to change my project idea because the code because to much and to hard for me.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think for me personally it is pretty obvious that the task that is going to require the most time and effort is the actual writing of the code. I think I have to implement the easiest parts of the code first like dataclasses and stuff that I feel confident will work and then slowly start to add components like the niche code for the prosthetics and other things slowly to make sure they do not make my whole code fall apart.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

I think that the distraction of my other school work because I still have a lot of exams coming up and still homework in other classes. So I have to give this project focus early on. I will minimize them by making sure I am ahead of all my work and I start things early. The resources I need available while working are just the ones that have been offered already like office hours and TA hours. Right now I do not have any questions but I do know where to go if questions come up.

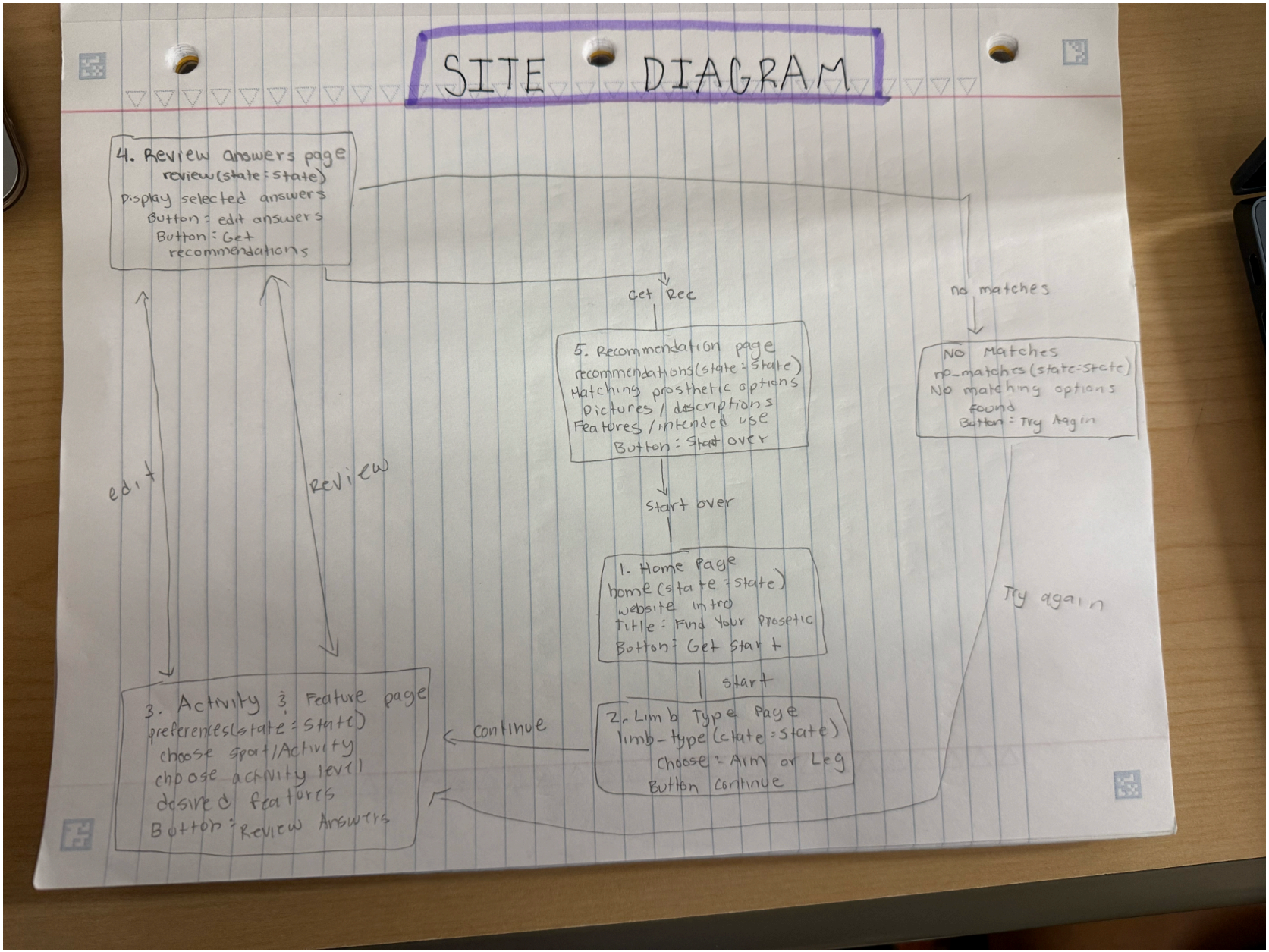
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


SITE DIAGRAM
4. Review answers page
review(state:state)
Display selected answers
Button: edit answers
Button: Get recommendations
edit
review
3. Activity & Feature page
preferences(state:state)
choose Sport/Activity
choose activity level
desired features
Button: Review Answers
continue
2. Limb Type Page
limb_type(state:state)
Choose: Arm or Leg
Button continue
start
1. Home Page
home(state:state)
website intro
Title: Find Your Prosetic
Button: Get Start
start over
5. Recommendation page
recommendations(state:state)
Matching prosthetic options
pictures / descriptions
Features / intended use
Button: Start over
Get Rec
no matches
No Matches
no_matches(state:state)
No matching options found
Button: Try Again
Try again

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	My website will help users explore prosthetic limb options based on their preferred activities, activity level, limb type, and desired features. It will compare their answers with a collection of prosthetic options and display recommendations that match their preferences.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	1. limb_type(str): Stores whether the user needs an arm or leg prosthetic. Updated when they make their selection. 2. activity(str): Stores the user's preferred sport or activity. Updated when they select an activity. 3. activity_level(str)(I could possibly also make this an int) (im still deciding): Stores low, moderate, or high activity level. Updated when selected. 4. desired_features(list[str]): Stores features such as comfort, stability, flexibility, and durability. Updated when users check or uncheck features. 5. prosthetic_options(list[Prosthetic]): Stores the collection of prosthetic options that the website can compare against the user's answers.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Dataclass: Prosthetic 1. name(str): Name of the prosthetic option. 2. limb_type(str): Whether it is an arm or leg prosthetic. 3. activity(str):The sport or activity it is designed for. 4. activity_level(str): The activity level it supports. 5. features(list[str]): Features such as comfort, stability, flexibility, and durability.

	6. description(str): Short explanation of the prosthetic and its intended use.
Routes (minimum 7) The names of your route functions	1. home: Displays the welcome page and introduces the website. 2. limb_type: Allows users to select an arm or leg prosthetic. 3. preferences: Collects the user's sport, activity level, and desired features. 4. review: Displays the user's answers so they can check them before continuing. 5. recommendations: Displays prosthetic options that match the user's answers. 6. no_matches: Displays a message when no prosthetic options match. 7. about: Explains how the website works and reminds users that recommendations are educational, not medical advice.
Pages (minimum 5) A brief description of the pages in your site	1. Home page: Introduces the website and includes a button to get started. 2. Limb type page: Asks users whether they are looking for an arm or leg prosthetic. 3. Preferences page: Asks users about their preferred activity, activity level, and desired features. 4. Review page: Shows the user's answers and allows them to go back and make changes. 5. Recommendations page: Displays matching prosthetic options with names, descriptions, and features. 6. No matches page: Explains when no prosthetic options match the user's selections and lets them try again. 7. About page: Explains the purpose of the website and how the recommendations are generated.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and	1. SelectBox(Arm or leg): Determines which type of prosthetic to recommend. 2. SelectBox(Preferred sport or activity): Helps match prosthetics to the user's interests. 3. SelectBox(Activity level: low, moderate, or high): Helps narrow down the prosthetic

their purposes For more information on these input types, see Drafter Documentation.	options. 4. CheckBox(Comfort, stability, flexibility, durability):Allows users to select the features they want. 5. TextArea(Optional additional preferences): Allows users to explain anything else they want considered.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1. If the user's limb type matches the prosthetic's limb type. a. Keeps only arm or leg options that match the user. 2. If the user's activity matches the prosthetic's intended activity. a. Helps recommend options designed for the user's preferred activity. 3. If the user's activity level matches the prosthetic's supported activity level. a. Removes options that do not match the selected activity level. 4. If the prosthetic has the features selected by the user. a. Helps narrow down recommendations based on desired features. 5. If there are no matching prosthetics. a. Sends the user to the No Matches page instead of the Recommendations page.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	My program will use a for loop to go through the list of prosthetic options stored in the State dataclass. It will compare each prosthetic's information with the user's answers and add matching options to a separate recommendations list. This allows the website to check multiple prosthetics without writing the same code repeatedly.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

how can i think abstractly to add another page and route into my website about prosetics

Since your website is a prosthetic limb recommendation tool, I would think about abstraction by asking yourself:

What is another important thing a person looking for a prosthetic might need to know that isn't already covered by my website?

In computer science, abstraction means focusing on the important information and leaving out unnecessary details. You want your new page to have a purpose that supports your website's overall goal without making the project overly complicated.

Here are some ideas!

1. Ideas for another page and route

의족 장애인을 위한 의수 소켓 - 중국 인공 다리, 의료용 다리

Page: Prosthetic Parts and Features

Route: /prosthetic-features

Explain the different parts of a prosthetic limb and what they do.

Socket

Knee joint

Prosthetic foot

Materials and weight

Why it works: It helps users understand what makes prosthetics different.

Middle-aged adults with blocked leg arteries may face high risk of amputation | American Heart Association

Page: Daily Activities

Route: /daily-activities

Explain how different prosthetics may support different activities.

Walking

Running

Playing sports

Everyday activities

Why it works: It connects the user's lifestyle to the recommendation process.

Prosthetic Leg Selection Guide: Key Factors for Optimal Fit & Function

Page: Compare Prosthetics

Route: /compare

Allow users to look at different prosthetic options side by side.

Price range

Weight

Intended activities

Features

Why it works: It gives users more information before making a decision.

2. How to think abstractly about the new page

Think about the information your program needs to keep track of rather than every individual detail.

For example, if you choose the Daily Activities page, you could think about it like this:

Abstract concept

Information to track

User

Name or user preferences

Activity

Walking, running, sports

Prosthetic

Type of prosthetic

Recommendation

Prosthetic that matches selected activities

Notice how you are grouping information into categories instead of creating a separate piece of code for every possible activity.

3. How the new route fits into your website

Here is an example of how your website navigation could work:

Home Page

Introduction to the website

User Information

/user-info
Daily Activities

/daily-activities
Recommendations

/recommendations
Compare Prosthetics

/compare
Learn More

/prosthetic-features
4. A question to help you brainstorm
I want you to think about this before choosing your next page:

What is the main purpose of your website?

Is it primarily to:

A. Help people understand the different types of prosthetics.

B. Help people find a prosthetic that fits their lifestyle.

C. Help people compare prosthetics before choosing one.

D. Educate people about prosthetics in general.

Your answer will help determine what your next page should accomplish and how it connects to the other pages.

One thing I would recommend is choosing a page that adds a new purpose to your website rather than simply repeating information you already have.

is it okay if my code seems a bit repetivite at times

Yes! It is completely okay if your code seems a little repetitive, especially since you're working on an introductory programming project using Drafter.

In fact, some repetition is normal when you're first learning how to build websites and organize code.

There is a computer science concept called abstraction that helps reduce repetition by grouping similar tasks together into reusable functions.

When is repetition okay?

- When two pages have similar layouts but display different information.
- When you repeat small pieces of code to make your website easier to understand.
- When you're still learning how routes and functions work.

When should you consider reducing repetition?

- When you copy and paste a large section of code multiple times.
- When you have to make the same change in several different places.
- When multiple functions perform almost exactly the same task.

My advice for your project

Since this is an introductory programming assignment, I would prioritize:

1. Making sure your website works.
2. Making sure each route has a clear purpose.
3. Keeping your code readable.
4. Looking for opportunities to use abstraction when it makes sense.

Remember: Your goal isn't to write the shortest code possible. Your goal is to write code that works, makes sense, and demonstrates what you've learned.

If you want, you can share a section of your Drafter code, and I can help you identify where repetition is perfectly fine and where abstraction might help without rewriting your code for you.

can you give me an example of a blank site diagram with 5 different page

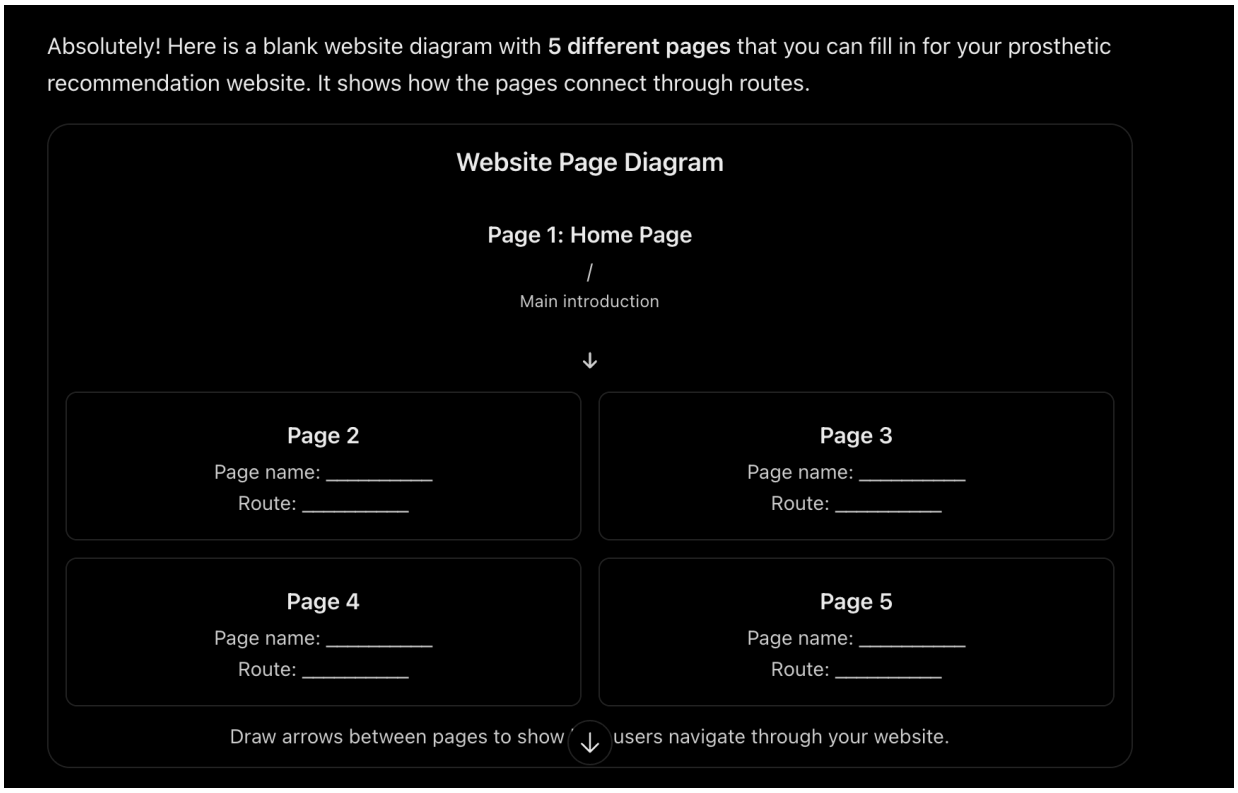
What you should include in each box

For each of your five pages, write:

- Page name: What the user sees.
- Route: The URL path (example: /recommendations).
- Purpose: What the user can do on that page.
- Navigation: Which other pages they can visit from there.

Tip: Your pages don't all have to connect in a straight line. Users should be able to navigate between pages using buttons or a navigation bar.

You can use this as a starting point for your assignment mockup!



Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.